

Doctoral Thesis Review

Name of the student: Jiří Klepl

Thesis outline title: Optimizing Memory Layouts and Traversal Orders in High-Performance Computing

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1. Relevance of the Research Topic

The topic of the dissertation is highly relevant and very timely for contemporary high-performance computing. Modern HPC systems are increasingly heterogeneous, and achieving high performance requires careful control over memory layout, traversal order, parallelization, and communication. At the same time, maintaining separate highly tuned implementations for different architectures is costly and error-prone. The thesis addresses this problem directly through layout- and traversal-agnostic programming.

Although related concepts exist in frameworks such as Kokkos, RAJA, or `std::mdspan`, the proposed Noarr approach is attractive because it remains lightweight, is implemented in native C++, and combines composable data-layout and traversal transformations while keeping them separate from the core algorithmic logic. The reported performance remains comparable to hand-written implementations.

I consider the integration with autotuning highly useful, and the MPI extension particularly valuable, since it allows local data layouts to change without modifying the communication logic, which can substantially simplify optimization of distributed and heterogeneous applications.

The LLM part is also timely, but more exploratory. Its conclusions are inherently less stable because LLM capabilities evolve very rapidly. Nevertheless, the finding that abstractions beneficial to human programmers may actually hinder LLM-generated optimized code is interesting and opens an important direction for future research.

2. Methods and Procedures

The methods and experimental procedures used in the thesis are generally appropriate and convincing. The proposed abstractions are evaluated on a representative set of applications and architectures, with performance systematically compared against conventional or hand-optimized implementations. The evaluation covers several important aspects of HPC programming, including shared-memory parallelization using OpenMP and TBB, CUDA execution, autotuning, and distributed-memory processing using MPI.

The inclusion of Cellato is somewhat less directly connected to the main Noarr development, since it represents a separate domain-specific library. Nevertheless, it applies the same underlying principle of separating computational logic from data layout and traversal and can therefore be regarded as a useful demonstration of the generality of the proposed approach.

One aspect that I find less extensively evaluated is CPU vectorization. While SIMD execution and compiler vectorization are discussed, there is limited analysis of how the proposed abstractions affect automatic vectorization or how explicit mechanisms such as OpenMP SIMD could be incorporated. A more detailed treatment of this issue would further strengthen the evaluation of the approach on modern CPUs.

3. Results and New Scientific Contributions

The dissertation presents several valuable and original scientific contributions. The layout-agnostic abstraction provides the foundation of the work and was developed collaboratively; nevertheless, the author declares a substantial contribution to its analysis and implementation. Particularly strong is the subsequent traversal-agnostic concept, including traversers and composable transformations, which extends the abstraction from data representation to execution order and parallelization. The results demonstrate that this can be achieved without significant performance loss compared with conventional implementations.

I consider the extension to distributed-memory computing using MPI one of the strongest contributions of the thesis. Automatically constructing MPI datatypes from layout abstractions and allowing communication logic to remain independent of the physical data layout is both original and practically useful, especially for heterogeneous systems.

The LLM-oriented research provides additional interesting findings, particularly that human-oriented abstractions may hinder rather than assist LLM optimization and that prompt specificity can strongly influence the resulting performance.

4. Quality of Formal Presentation

I consider the formal presentation the weakest aspect of the dissertation. Although the individual appended publications are detailed, the introductory thesis text is surprisingly brief and high-level and does not form a sufficiently self-contained scientific narrative. The introduction states two broad research directions, but does not formulate clear overarching hypotheses, research questions, partial objectives, or an explicit list of scientific contributions and corresponding results.

The presentation of results is particularly unsatisfactory. Apart from several simple illustrative code listings, the synthesis contains essentially no figures and very little quantitative evidence or systematic comparison. Important performance results, methodological details, and experimental evidence therefore have to be searched for in the individual appended papers. In my view, a dissertation should synthesize these results and guide the reader through the scientific argument rather than require reconstruction of that argument from a collection of publications.

A further serious deficiency is that the author's individual contribution to the joint publications is presented only after the complete thesis summary, in the List of Publications. This information should have been provided at the beginning and explicitly linked to the individual thesis contributions. Overall, I find the formal presentation significantly below the standard I would expect from a doctoral dissertation and difficult to accept in its present form.

5. Defense Questions

- Are you aware of any research groups or developers outside your department who currently use Noarr in their own applications? If so, what types of applications are they using it for? If not, what do you see as the main obstacle to wider adoption of the library?
- Your experiments indicate that abstractions designed to help human programmers, including Noarr, Halide, and Exo, can actually hinder LLMs when generating optimized code. What do you think an abstraction designed specifically for an LLM should look like? Would it still resemble a high-level programming abstraction, or would a more explicit intermediate representation exposing dependencies, memory access patterns, parallelism, and hardware constraints be more suitable?
- In the MPI datatype construction, why did you choose an MPI_Type_create_hvector-style representation with strides expressed in bytes rather than MPI_Type_vector, where the stride is expressed in multiples of the extent of the underlying datatype? For regular layouts, the latter would appear simpler and avoid explicit byte-stride calculations. Is hvector necessary to preserve the general composability of arbitrary Noarr layouts, or were there other reasons for this design decision?
- You demonstrate parallel execution using OpenMP, TBB and CUDA, but SIMD vectorization on CPUs is discussed much less extensively. How do Noarr abstractions affect compiler auto-vectorization? Can the abstraction expose sufficient information to the compiler, and could explicit mechanisms such as `#pragma omp simd` or SIMD-aware traversers be naturally incorporated?
- Cellato is implemented as a separate domain-specific library rather than as an extension or application of Noarr. Why was it necessary to create a new abstraction instead of expressing the cellular-automata optimizations directly using Noarr? What did you learn from Cellato that could not have been demonstrated using Noarr itself?

6. Conclusions

Overall, the thesis presents a valuable and timely contribution to the field of high-performance computing, particularly in the area of layout- and traversal-agnostic programming, performance portability, and distributed-memory execution. I consider the traversal abstractions and especially the MPI extension among the strongest results, while the work on LLM-assisted optimization provides several interesting and relevant observations for future research.

The scientific results are supported by a solid publication record in respected conferences and peer-reviewed journals, demonstrating the author's ability to conduct high-quality research and contribute to several areas of HPC. The Noarr library represents a useful and technically interesting software outcome with potential for broader adoption.

At the same time, I consider the formal presentation of the dissertation a significant weakness. The introductory synthesis is too brief and high-level, and important objectives, individual contributions, quantitative results, and methodological details are often accessible only through the appended publications. This makes the thesis unnecessarily difficult to evaluate as a coherent scientific work.

Despite these reservations regarding presentation, I consider the scientific contributions sufficient and valuable, and I recommend the thesis for defense and, subject to a successful defense, for the awarding of the PhD degree.

In Brno, 2nd September 2026

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prof. Ing. Jiří Jaroš, Ph.D.